Lab 4 – Vibe Coding

1. Project Setup

User: The lab requires running the assigned Python/Pygame Coin Collector, recording a 10-second video before changes, using Vibe Coding prompts to fix the broken code and add README features, recording a second video after changes, and making separate commits for each task.

Assistant: The working project was located under Lab-4/coin-collector. The project was run with python main.py. The original game was to be recorded before modifications.

2. Task 1 – Fix Coin Collection Bug

User prompt: Inspect the existing Python Pygame Coin Collector project. Fix only the coin collection bug. When the player touches a coin, its value is added every frame because the collected coin remains in the coin list. A coin must be collected exactly once and then removed from the active coin list. Make the smallest possible change and do not modify unrelated functionality.

Implementation: In game_engine.py, after adding the collected coin's value to the score, remove that coin from the active list.

```
for coin in collected:
self.score += coin.value
self.coins.remove(coin)
```

Test: Touching one coin increases the score once and the coin disappears. The user confirmed the fix worked.

Commit: Fix coin collection bug

3. Task 2 – Three Coin Types

User prompt: Implement only Task 2. Add at least three visually distinguishable coin types: Bronze = 1 point, Silver = 3 points, Gold = 5 points. Reuse the existing Coin class and project structure. Keep the coin collection fix working and do not add obstacles, lives, timers, game-over, or restart yet.

Implementation: Reuse the existing Coin class and select a random type with the required value and color.

```
COIN_TYPES = [
{"value": 1, "color": (205, 127, 50)}, # Bronze
{"value": 3, "color": (192, 192, 192)}, # Silver
{"value": 5, "color": (255, 215, 0)}, # Gold
]
```

Test: Bronze gives +1, Silver gives +3, Gold gives +5, and each collected coin disappears and scores only once.

Commit: Add multiple coin types

4. Task 3 – Obstacles and Lives

User prompt: Implement only Task 3. Add multiple visible obstacles, start with 3 lives, display the lives, decrease one life on obstacle collision, and prevent continuous collision from decreasing multiple lives every frame. Keep all coin types and the coin collection fix working. Do not add timer, game-over, or restart yet.

Implementation: Add obstacle rectangles, lives = 3, and an obstacle_hit state. A life is removed only when collision begins; the state resets after the player moves away.

```
if touching_obstacle and not self.obstacle_hit:
self.lives -= 1
self.obstacle_hit = True

if not touching_obstacle:
```

```
self.obstacle_hit = False
```

Test: Start with 3 lives. A collision reduces one life. Staying against an obstacle does not rapidly drain lives; moving away and colliding again permits another life loss.

Commit: Add obstacles and lives

5. Task 4 – Timer, Game Over and Restart

User prompt: Implement the final feature. Add a 30-second countdown using real elapsed time. End the game when time reaches zero or lives reach zero. Display GAME OVER and the final score. Press R to restart. Restart resets score, lives to 3, timer to 30, coins, obstacles, and player position. Keep all previous features working.

Implementation: Add a 30-second timer, a game-over state, and a restart method. Gameplay stops after game over except for the R key.

```
elapsed_seconds = (pygame.time.get_ticks() - self.start_time) // 1000
self.time_left = max(0, GAME_DURATION - elapsed_seconds)

if self.time_left <= 0 or self.lives <= 0:
self.game_over = True
```

Restart: Press R after game over to reset the player, coins, score, lives, obstacles, timer, and game-over state.

Test: Confirm the timer counts down from 30, collisions reduce lives, coin values remain correct, GAME OVER appears at zero time or zero lives, and R starts a fresh game.

Commit: Add timer game over and restart

6. Final Submission Checklist

1. Before video – approximately 10 seconds showing the original game/bug.
2. After video – approximately 10 seconds showing the completed game.
3. Updated Coin Collector source code.
4. chat history PDF.

Final commits:
• Fix coin collection bug
• Add multiple coin types
• Add obstacles and lives
• Add timer game over and restart

Repository requirement: Push the work to the personal lab repository. Do not raise a PR to the SETAPESU26 main repository.